1.0 Introduction

Music plays a significant role for many people while they complete tasks in daily life, especially when focusing on an activity, like typing. Tasks completed online, such as research or completing official documents, must involve typing on a computer or laptop [1]. This proposal provides a variety of aspects of the research question: ***"Does music as a distraction type, along with text familiarity, affect typing speed and accuracy?"*** As music is used frequently when typing, the initial hypothesis is that it is helpful. The team wishes to examine the credibility of this claim with collected statistical data, along with the hypotheses. Thus, the goal of the experiment based on this research question is to verify if music, in addition to text familiarity, acts as a tool to increase efficiency when completing tasks (for instance, lecture notes for students compared to articles that individuals have not seen before). Referring to the literature review (detailed in Section 2.0), the design experiment will be based on the results reviewed in that literature. Further details are discussed in Section 3.0, along with statistical analyses on the data collected to answer the stated research question, as outlined in Section 4.0 and beyond.

2.0 Background

Typing speed and accuracy are measures of typing performance that can be influenced by factors like environmental conditions and cognitive distractions. There are mixed reviews on the idea of listening to music while typing, as some individuals claim that it helps them focus, whereas others find it a distraction. The volume or the type of music (instrumental or vocal music) can play a role in how much impact it has on cognitive ability. Additionally, text familiarity may also contribute to the speed and accuracy with which an individual can type. This study seeks to examine how the factors of music and text familiarity (e.g., cues and free recall) can influence typing speed and accuracy, to determine the overall effect on typing performance.

2.1 The Impact of Music on Typing Performance

The effect of background music on cognitive tasks has been a subject of interest in psychological research. Bramwell-Dicks, Petrie, and Edwards (2016) conducted a study to analyze what factors (e.g., type of music and volume) and individual characteristics (e.g., typing ability) may contribute to a change in cognitive ability. Their findings suggest that the presence of music—particularly music with vocals—can negatively affect typing performance, as participants typed more slowly and made more errors, compared to when less vocal music was played. This finding supports the idea that music, as a potential distraction, can interfere with regular tasks requiring concentration and accuracy, such as typing.

Music plays a significant role in defining typing performance, especially when considering variables like vocal presence, music volume, and skill level for typing [2]. The presence of vocals in the music introduces a level of distraction that impairs cognitive focus, making the typing task more difficult [2]. The suggestion was that vocal music likely competes for cognitive resources that would otherwise be used for the task at hand (in this case, typing), reducing the overall efficiency of said task. The study also found that the impact of vocal music differed based on the typists' abilities, yet even faster typists were noticeably affected by vocals, as their typing speed and accuracy reduced when compared to conditions when only instrumental music was played. A second study in the same research also concluded that "high volume music can have a negative [e]ffect on typing speed," helping to hypothesize that louder and more vocal music tends to have a more negative impact on cognitive ability than soft and instrumental music [2].

2.2 The Impact of Text Familiarity on Typing Performance

The familiarity of a given text impacts cognitive processing since it enhances recall efficiency and memory performance, therefore influencing both cued and free recall. According to Tulving and Pearlstone (1965), cued recall improves when retrieval cues are available, but its efficiency decreases as the number of items per cue increases [3]. Free recall relies on prior exposure to text for better recall [3]. Similarly, Miller's (1956) theory suggests that chunking can enhance recall when familiar information is grouped into specific chunks that are related to one another [3].

However, Mathews (1954) found that when too many items are linked together into a single cue, it limits the benefits of familiarity as recall performance declines [3]. This has direct implications for typing performance, as typing familiar texts can increase speed and accuracy due to the ability to anticipate the upcoming words, but an overreliance can also create errors when there are too many cues [3]. The results from the cued and free recall studies suggest that cued recall performance significantly varied based on the number of items per cue, while free recall performance remained more consistent regardless of the information chunks [3].

Hence, excessive reliance on chunked information can reduce performance in terms of retrieval accuracy when too many words are assigned to a single cue [3]. Overall, these findings suggest that typing speed and accuracy should improve given familiar texts since prior familiarity can make it easier for participants to recall and process words more quickly. Yet, the right balance of familiarity needs to be found to avoid an overreliance on structured chunking of information that can decrease speed or accuracy in such a way that makes it more difficult to adjust when unexpected wording is encountered.

3.0 Proposed Methods

Since participants have different typing characteristics, it is necessary to record data for individuals' performances. Thus, two types of data from each participant will be collected for this project: typing speed and accuracy before and during the experiment (words per minute and error percentage, respectively). The team will measure participants' speed and count errors using Monkeytype [4], both before and after listening to different types of music. As a result, there will be six pieces of data collected for each participant: typing speed and accuracy before the experiment, when listening to slow-paced music, and fast-paced music.

3.1 Independent and Dependent Variables

The first independent variable will be the level of distraction, which is broken down into three categories: no music, slow-paced music, and fast-paced music. Music types are defined through tempo, measured in beats per minute (BPM). For this study, slow-paced music will be Goodbye in Her Eyes by Zac Brown Band, 81 BPM, while fast-paced music will be Birthday by the Beatles, which is 140 BPM. The Dave Tompkins Music Database [5] is used to select songs for the study based on BPM. The second independent variable will be text familiarity. This refers to whether the participants have prior exposure and comprehension of the content of a given text that they will type, or if the text is entirely new. To keep the text length consistent, all text will be 1000 characters long. The familiar text will be the excerpts from the MIE240 course's case studies and will be verified by confirming the completion of the case study assignment submission. The unfamiliar text will be randomly generated words using MonkeyType [5] to ensure the writing material is equally unfamiliar to all participants. The dependent variables will be the speed and accuracy of the participants' typing performances on a given text. The speed will be measured through words per minute (WPM). The accuracy will be assessed through a percentage representation of correctly typed characters against the total characters. Monkeytype [4] will be used to measure WPM and accuracy percentage. The participants' WPM of no music and no text familiarity will be used as an independent variable, representing the participants' typing skills with no external factors.

3.2 Participants

The study will involve post-secondary Industrial Engineering students, aged 19 to 24, with typing experience to ensure familiarity with the selected material (MIE240 case studies). A total of twelve students will be included, providing sufficient data for statistical analysis.

3.3 Hypotheses and Alpha (α) value

The following are the null and alternative hypotheses for the project:

H_0: Music and text familiarity have no effect on typing speed and accuracy.

H_1: Music and text familiarity have an effect on typing speed and accuracy.

The team will use an alpha value of **0.05** for the statistical tests and analyses, as a lower alpha value increases the likelihood of rejecting the null hypothesis and keeps a balance between minimizing errors. It is also the conventional threshold to make a strong claim.

3.4 Experiment Procedures

The participants will be located in an isolated area, with no other devices or nearby individuals to avoid external distractions that could potentially influence the data collected. The experiment will start with the base trial (trial 0), to measure the speed and accuracy of the participants' typing under the two conditions of no music and no text familiarity. Including the base trial, participants will complete six experimental trials:

1. No music and no text familiarity (base case)
2. No music and text familiarity
3. Slow-paced music and no text familiarity
4. Fast-paced music and no text familiarity
5. Slow-paced music and text familiarity
6. Fast-paced music and text familiarity

Participants will be given a 10-minute break between trials to prevent fatigue from being factored into the results. After recording the speed (in WPM) and accuracy (in percentage) from each trial, the results will be compared with trial 0's results. The $+ / -$ values for the change in WPM and accuracy percentage will be used as the final data for subsequent statistical analysis. To minimize external influences, the music volume will remain at 50% on the phone regardless of pace, and noise-cancelling headphones will be used for a no-music environment. The experiment will be conducted in ECF labs (e.g., MB123 or GB144) to ensure keyboard consistency and familiarity. To control for text familiarity, participants will review case study excerpts before typing, and the trials will start with the base case (no music, no text familiarity) before following the specified trial sequence to avoid confusion from abrupt text changes.

4.0 Results

Based on collected data regarding WPM and accuracy, an adjusted WPM was used as a standardized metric in place of individual raw values. This new WPM was used below as the basis of overall data analysis. The experimental results were analyzed using both descriptive and inferential statistics. Graphs generated using R code illustrate the impact of music pace and text familiarity on typing speed and accuracy. The following section identifies observed trends and their significance through statistical tests and summarizes key findings. Table 1 presents essential statistics for analysis.

Table 1. Summary and two-way ANOVA statistics.

Summary

```
> summary(model)

Call:
lm(formula = Results ~ Music.Pace + Text.Familiarity, data = data)

Residuals:
     Min       1Q   Median       3Q      Max 
-12.8958  -2.9375   0.6875   3.4063  10.6875 

Coefficients:
                 Estimate Std. Error t value Pr(>|t|)  
(Intercept)      -2.17055    3.42139  -0.634   0.5290  
Music.Pace        0.01059    0.02897   0.366   0.7163  
Text.Familiarity -3.79167    1.70906  -2.219   0.0316 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5.92 on 45 degrees of freedom
Multiple R-squared:  0.101,	Adjusted R-squared:  0.06105 
F-statistic: 2.528 on 2 and 45 DF,  p-value: 0.09111
```

Anova

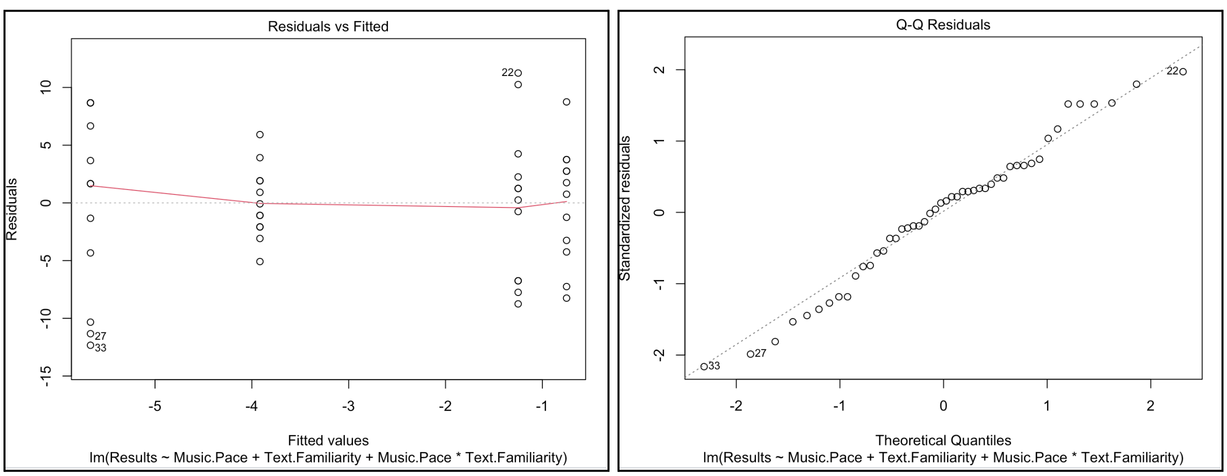
```
> res.aov1 <- aov(Results ~ Music.Pace, data = data1)
> summary(res.aov1)
            Df Sum Sq Mean Sq F value Pr(>F)
Music.Pace   1   18.4   18.38   0.656  0.427
Residuals   22  616.6   28.03               
> 
> res.aov2 <- aov(Results ~ Music.Pace, data = data2)
> summary(res.aov2)
            Df Sum Sq Mean Sq F value Pr(>F)
Music.Pace   1    1.5    1.50   0.043  0.837
Residuals   22  760.5   34.57               
> 
> res.aov3 <- aov(Results ~ Music.Pace + Text.Familiarity, data = data3)
> summary(res.aov3)
                 Df Sum Sq Mean Sq F value Pr(>F)  
Music.Pace        1    4.7    4.69   0.134 0.7163  
Text.Familiarity  1  172.5  172.52   4.922 0.0316 *
Residuals        45 1577.3   35.05                 
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

4.1 Analysis and Discussion on Model Assumptions

Before any further analysis of the data and hypothesis, the following assumptions were verified to see if they were satisfied:

1. Independent errors: For any two observations, the residual terms should be uncorrelated (or independent). A *Durbin-Watson* test was applied to the residuals to assess the independence of errors in the team's model. After extracting the wavelet coefficients, a t-test on the null hypothesis of 'mean of the wavelet coefficients is zero' was performed. The test yielded a p-value of *0.3912*, a number greater than the alpha value of 0.05. As a result, the null hypothesis was rejected, confirming that the assumption was satisfied.

2. Normally distributed errors: The residuals in the model appear to be normally distributed random variables with a mean of 0. To make sure that the residuals of the model were normally distributed, the graphs were created. The Residuals vs. Fitted graph (Figure 1) shows a random scatter of dots around zero, with no discernible curve. The Q-Q plot (Figure 2) shows no clear pattern, with data crossing the centerline multiple times, similar to a normal distribution. Thus, the residuals appear to be normally distributed.



Figures 1 and 2. Residuals vs. Fitted and Q-Q Plot to test the normality of residuals

3. Homoscedasticity (homogeneity of variances): The residuals should exhibit constant variance across all levels of the predictor variable(s). The Scale-Location plot (Figure 3) displays a

random scatter of points around the red line with a relatively uniform spread across the range of fitted values, indicating that the residuals are homoscedastic.

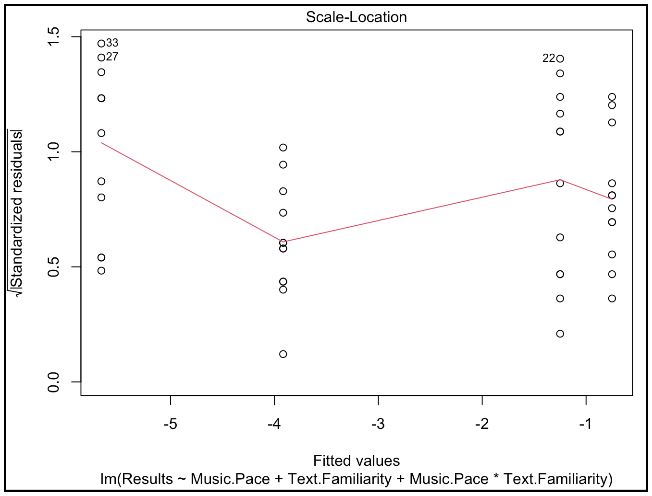


Figure 3. Scale-Location plot to test Homoscedasticity

4. Lack of perfect multicollinearity: Predictors should not show perfect linear relationships. To check for the assumption of multicollinearity, pairwise correlation coefficients of any 2 predictors were plotted. As shown in the correlation plots (Figure 4), no coefficients are equal to 1 or -1, indicating that there is no perfect multicollinearity among the predictors.

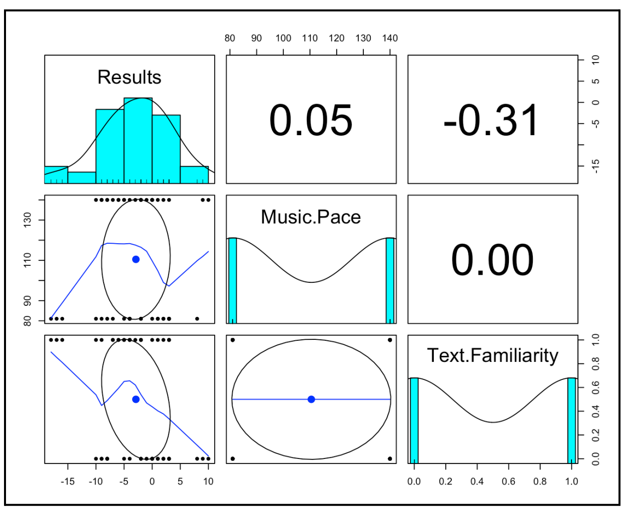
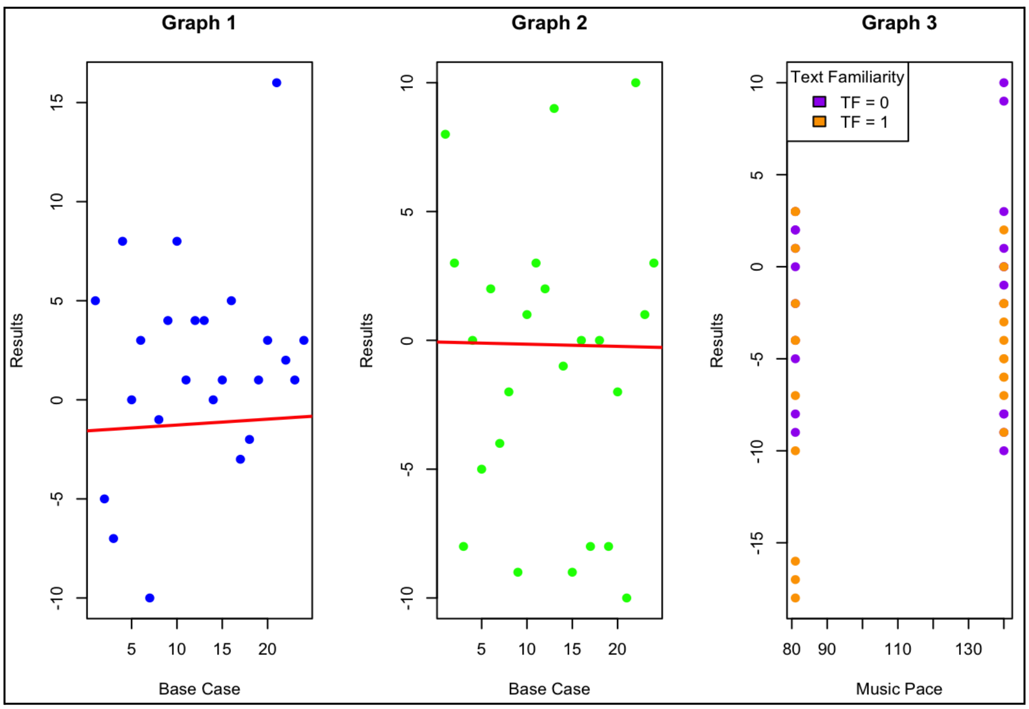


Figure 4. Correlation plots for any 2 predictors in the dataset.

All the tests and visuals tested for assumptions gave satisfactory results, allowing the team to move on to the actual analysis of the data, both descriptive and inferential.

4.2 Descriptive Statistics of the Data

Descriptive statistics identified general trends in typing speed and accuracy across six trials under different conditions. Participants generally typed faster and more accurately with familiar texts, particularly when paired with slow-paced music, which showed the greatest improvement. Slow-paced music appeared to slightly improve typing performance, while fast-paced music introduced more variation in typing results, particularly with unfamiliar texts, where performance tended to decline. Although both text familiarity and music pace influenced typing outcomes, their combined effect was not statistically significant. These observed trends provided a foundation for further statistical testing, as the significance of these effects was not confirmed.



Graph 1:
Data: trials 5 and 6
Base case: trial 2

Graph 2:
Data: trials 3 and 4
Base case: trial 1

Graph 3:
Data: trials 2, 3, 4, 5 and 6
Base case: trial 1

Figure 5. Effects of music pace and/or text familiarity on typing performance.

The three graphs in Figure 5 visually represent trends from the experiment. Graph 1 displays the results for different music paces with familiar text, while Graph 2 displays the same comparison but for unfamiliar texts. The slopes in both Graphs 1 and 2 are relatively flat, indicating that neither condition alone strongly influenced typing performance. However, Graph 3 combines both variables, showing a visible upward shift in the results for familiar text (TF = 1, where TF means text familiarity) compared to unfamiliar text (TF = 0). This suggests that while the interaction between music pace and text familiarity may contribute to small differences in typing performance, the overall effect was not significant. Therefore, the graphs in Figure 3 support the conclusion that both music pace and text familiarity had a minimal impact on typing performance, aligning with the findings in Section 4.3 below.

4.3 Inferential Statistics of the Data

Inferential statistics were conducted to evaluate the significance of the collected data and to statistically determine whether music and text familiarity significantly influence typing speed. The analysis began by defining the null and alternative hypotheses (Section 3.3) and testing them through data simulation using R, generating two-way ANOVA tables. They compare typing performance across different conditions: presence or absence of music and text familiarity. The ANOVA tests returned F-statistics to determine whether observed differences were statistically significant or due to random chance. The p-values for the three cases are as follows:

1. Music vs. familiar text: $p = 0.427$
2. Music vs. unfamiliar text: $p = 0.837$
3. Music vs. both familiar and unfamiliar text: $p = 0.7163$

All p-values exceeded the conventional significance threshold of the chosen alpha value of 0.05. In addition, the third linear model, incorporating dummy variables for text familiarity and music paces, resulted in a low adjusted R^2 value (~ 0.05), indicating that the model can not explain much about the variability of the dependent variable (typing speed). Given the high p-values and low adjusted R^2 value (Table 1), the null hypothesis failed to be rejected with a high degree of confidence, providing strong statistical evidence that music and text familiarity do not significantly affect typing speed.

4.4 Conclusions of Data Results

The experiment addressed the research question: **"*Does music as a distraction type, along with text familiarity, affect typing speed and accuracy?*"** Descriptive statistics showed that participants generally typed faster and more accurately when typing familiar texts when listening to slow-paced music but with a small impact, whereas fast-paced music introduced more variability with unfamiliar texts. Although it suggested that both music pace and text familiarity affect typing speed and accuracy, their interaction did not yield statistically significant results, supported by the graphs in Figure 5, showing only minimal shifts in performance across different conditions. Additionally, inferential statistics using two-way ANOVA and linear modelling confirmed the lack of statistical significance in the effects of music and text familiarity on typing speed and accuracy. All p-values (as detailed in Section 4.3) exceeded the alpha value of 0.05, and the low adjusted R^2 indicated a weak representation of the models. Thus, the null hypothesis was not rejected, providing strong evidence that these variables have a negligible impact on typing performance.

5.0 Discussion

The following section discusses the meaning of the results of this experiment to address the stated research question. It also outlines the experiment's strengths, limitations, and potential opportunities for future research.

5.1 Implications of Results and Experiment

The conclusions from the descriptive and inferential statistics have failed to directly answer the research question, as stated in Section 1.0, as neither music nor text familiarity exhibited a significant linear relationship with the accuracy-adjusted typing speed. In fact, raw data showed that slow-paced music and text familiarity (Trial 5) yielded the highest WPM, directly rejecting the claims based on the literature reviews that fast-paced music and text familiarity affect typing speed and accuracy. Instead, the results discovered other key components that more directly affect typing performance, such as text difficulty, the number (or presence) of quotations, special characters, and capital letters. The team concluded that typing is largely a subconscious process rather than a full cognitive action. Individuals simply look at character after character on simple typing replications (rather than creative typing), making them less susceptible to divided attention from external variables such as the independent variables.

The experiment results provided insight into the effects of music and text familiarity on overall typing performance. While previous literature studies suggested that music, especially fast-paced music, could negatively impact typing performance due to reduced cognitive focus, this experiment did not find statistically significant effects of either music or text familiarity on typing speed and accuracy. Some patterns were observed, such as slower typing speed with fast-paced music or improved typing outcomes with familiar texts, but these trends were not strong enough to reject the null hypothesis. In addition, familiar texts also contributed to improved typing outcomes, which was likely due to reducing errors and allowing the participants to possibly anticipate upcoming words. Therefore, the results suggest that typing tasks can be influenced by other conditions, such as music and text familiarity, and not just individual abilities that can include typing experience or attention span. Overall, while individual differences still played a role, the findings of this experiment do not confirm that both music pace and text familiarity are meaningful variables in assessing typing performance. Hence, these outcomes do not support the general claim (i.e., hypothesis), but rather highlight areas that would benefit from further research, as outlined in Section 5.4 below.

5.3 Experiment Strengths and Limitations

The strength of the experiment was its controlled environment, which minimized unexpected distractions. Additionally, the procedure is easily replicable, allowing other researchers to verify the results and enhance reliability. Further refinement with stricter constraints could better explore the true relationship between music and text familiarity, and the pace of the music that the participants listen to, providing deeper insights into the research question. However, a limitation was the presence of unfamiliarity with the keyboards. Although the experiment was conducted in ECF labs to maintain consistency, some participants improved their typing performance over time as they adjusted to the lab keyboards. Text difficulty was also not accounted for, though all texts were limited to 1000 characters for consistency. Participants typed slower and made more errors when encountering long or uncommon words (e.g., greater than 10 characters or words like "ecstatic"), affecting both WPM and accuracy. These unintended influences introduced challenges in fully isolating the intended experimental factors.

5.4 Potential Future Experiments

Supervision of the experiments for this study suggested room for improvement if replication were to take place, as flaws in the current structure were better highlighted during the testing phase. Ways to resolve some of these flaws include:

1. Increase the sample size of participants and diversify the demographic.

a. By conducting the experiments with an increased number and demographic diversity of participants, the reliability of the findings and the confidence of the statistical analysis would be improved. The current study's limited sample size and lack of variability in participant backgrounds could have hindered the ability of the experiments to detect any significant results or reveal hidden trends.

2. Shorten the passage size.
    a. Another observation from the experiments was that participants experienced fatigue and wrist strain from typing 1000 characters per trial, despite taking a break between trials. In a future run, shortening the passage size to 500 characters can ensure meaningful testing while considering the gradual physical effects on the participants.

While this study provided valuable insights into the influence of music and text familiarity on typing performance, other avenues for future research could clarify the background context:

1. Investigate various music genres and tempos.
    a. The genre of music (e.g., classical, upbeat pop, instrumental) may influence typing performance. The current study indicated that a change in BPM affected typing performance, propelling the idea that some genres could be more conducive to concentration and performance than others. Based on the statistical results, a new hypothesis could be that more upbeat genres with higher BPM may hinder an individual's typing performance and task complexity.
2. Explore the relation between text complexity and cognitive load.
    b. Since the current study was based on familiarity with the text, the experiments were not suited to test whether the complexity of the text also had an effect on the overall performance. Supervision during the experiments suggested that individuals may find changes in their cognitive load due to varying punctuation, sentence lengths, and/or vocabulary difficulty, setting up a potential hypothesis for a future experiment.

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